# Yifan Yuan

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## EDUCATIONAL BACKGROUND

**Carnegie Mellon University, School of Computer Science** **Pittsburgh, United States**
Master of Science in Intelligent Information System May.2028

**The University of Melbourne** **Melbourne, Australia**
*Bachelor of Science in Data Science; GPA: 3.917/4.0* Dec. 2025

**Exchange Program in University of Michigan, Ann Arbor** **Ann Arbor, United States**
*College of Engineering; GPA: 3.97 / 4.0* Aug. 2024 – May.2025

- **Key Modules:** Machine Learning, Statistics, Stochastic Process, Optimization (Non-linear)

## INTERNSHIP EXPERIENCES

**Shanghai Electric** **Shanghai, China**
*AI Product Development Intern* Feb 2026 - Jul. 2026

- Developed an end-to-end recruitment agent for complex candidate–role matching, combining structured competency analysis with human-in-loop for traceable screening.
- Translated ambiguous needs into an evolving talent intelligence network prototype, where existing employee profiles and potential candidate pools are continuously enriched and matched for emerging workforce needs to support internal-first mobility and external talent supplementation.
- Built a flexible-manufacturing industrial agent and showcased it as an applied AI case for AI adoption in manufacturing.

**Schneider Electric** **Shanghai, China**
*Machine Learning Intern* May 2025 - Jul. 2025

- Initiated a multi-agent system in LangGraph for virtual factory operations and led company-wide workshops to support its adoption.
- Built a Python-based statistical modeling system with SQL integration for real-time production anomaly detection from 0 to 1, achieving 97.1% accuracy.
- Developed case-based AI vision tutorials for Lighthouse Factories, facilitating industry–academia integration.

**Center for Healthcare Engineering & Patient Safety (CHEPS), University of Michigan** **Ann Arbor, United States**
*Research Assistant, Reimagining Operations for Antenatal Care Delivery Team* Jan. 2025 – May. 2025

- Orchestrated a data-processing pipeline through case-study-based data cleaning, feature engineering, and cross-dataset integration to estimate tailored policy application in current healthcare system.
- Collaborated with clinicians to trace data back to clinical workflows, validating a reliable dataset for downstream analysis.
- Contributed to the methodology section of a clinical research paper and quantified operational and clinical impacts of policy.

**Aptiv (Delphi)** **Shanghai, China**
*Data Analysis Intern* Nov. 2023 - Feb. 2024

- Participated in the preliminary design of a big data platform and was responsible for its data processing, extracting and processing millions of daily records using SQL for analysis.
- Built interactive dashboards in Tableau to visualize dynamic data, such as energy consumption over normal time based on different subdivision levels and time windows.

## RESEARCH EXPERIENCES

**Buy-Now-Pay-Later (BNPL) Merchant Risk Modeling & Ranking System** Aug. 2025 - Nov. 2025

- Built an end-to-end ETL pipeline in Python consolidating multiple internal and external datasets into analysis-ready schema.
- Designed a custom KNN "look-alike" fraud model for extreme label sparsity (114 fraud merchants, ~13M transactions), optimizing feature weights to distinguish fraudulent merchants and propagate fraud risk.
- Engineered consumer loyalty,revenue concentration(HHI),and growth indicators to identify resilient,high-potential merchants.
- Prototyped an interactive merchant platform combining KNN fraud risk with customizable indicator-weighted rankings.

**The Role of Multiplicative Noise in Cellular Decision-Making: Insights from SDEs** Jan. 2025 - Feb. 2025

- Applied stochastic differential equations (SDEs) in Julia to model noise-driven dynamics in complex systems, deriving steady-state probability distributions $P_s(x)$ and probabilistic landscapes $U_q(x)$.
- Measured how noise intensity shapes system behavior, such as driving transitions from monostable to bistable system.

**Optimizing Image Classification: Layer Freezing, and LoRA Adaptation** Oct. 2024 - Dec. 2024

- Optimized layer-wise parameter freezing on ResNet-34, improving accuracy from 75.9% to 90.6% over FC-only fine-tuning while using 38% fewer parameters than full fine-tuning.
- Adapted LoRA from Transformers to CNN architectures, identifying architectural constraints in ResNet and reducing trainable parameters by 99.7% (119.6M → 0.33M) on VGG-16 FC layers with only a 3.5 pp accuracy decrease.

## SKILLS

**Programming:** Python, R, C, Julia, VBA, SQL (DBeaver, MySQL Workbench)
**Tools & Platforms:** Excel, Tableau, PyTorch, PySpark, ProModel
**Languages:** Mandarin (Native), English (IELTS 7.5)